

PRANAV BALAKRISHNAN

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EDUCATION

University of Massachusetts Amherst – MS in Computer Science | CGPA 3.97/4.00 | May 2026

Indian Institute of Technology Madras – BS in Data Science | CGPA 8.31/10.00 | May 2024

PUBLICATIONS

1. “From Token Probabilities to Semantic Constraints: Towards Declarative Probabilistic Evaluation of Language Models” - Accepted to the Main Conference and Nominated for an Empirical Methods in Natural Language Processing (EMNLP) 2026 Award
2. “GPU-Accelerated Generalized Labeled Multi-Bernoulli Filter for Multi-Detection Point Targets” – Submitted to IEEE Transactions on Signal Processing (TSP) Journal 2026
3. “GPU-GLMB: Assessing the Scalability of GPU-Accelerated Multi-Hypothesis Tracking” - Accepted to IEEE Military Communications Conference (MILCOM) 2025
4. “IoBT-MAX: From Multimodal Data Collection to Distributed Real-Time Analytics Experimentation” - Submitted to IEEE Military Communications Conference (MILCOM) 2026
5. “Improving Target Tracking in Stone Soup via State-Dependent Process Noise Models” - Submitted to International Conference on Information Fusion 2026

RESEARCH & WORK EXPERIENCE

Allen Institute of AI (AI2) - Research Extern

Jan 2026 – Aug 2026

- Developed a novel framework for **evaluating and training LLMs** with logical consistency constraints, integrating probabilistic reasoning and neuro-symbolic methods to measure semantic consistency beyond standard language modeling objectives.
- Designed theoretical methods for selecting effective logical constraints, analyzing their information content and optimization behavior to identify constraints that provide strong and stable learning signals for LLM training.
- Built scalable training and evaluation pipelines for **open-source LLMs (OLMo, SmoLLM)**, conducting large-scale experiments that uncovered systematic reasoning and consistency failure modes while validating new constraint-based training approaches.

Army Research Laboratory (ARL) - Uncertainty Aware Perception and Multi-Object Tracking

Jun 2025 – Aug 2026

- Developed **MDPT-GLMB**, an uncertainty-aware **multi-sensor perception and multi-object tracking** framework combining **neural-network object detection** with structured **Bayesian inference** for real-time state estimation.
- Identified the **one-object–one-measurement assumption** and **sequential-scan Gibbs sampling** as key computational bottlenecks in standard GLMB, then introduced a **Multiple-Detection Point Target (MDPT)** model supporting multiple detections per object while preserving **GLMB closed-form conjugacy**.
- Designed an **Approximate Block Ancestral (ABA) sampler** to generate complete measurement-to-track association hypotheses in parallel, and built a highly optimized, fully vectorized **PyTorch/GPU implementation** evaluated on synthetic and real-world data with clutter, missed detections, blind spots, and complex vehicle trajectories.
- Achieved **8-20× faster filter updates** with nearly identical tracking accuracy and **sub-100 ms latency** across Dockerized **Jetson Orin, RTX 4090, and NVIDIA L40S** platforms, enabling real-time edge-to-server deployment.

Improving Spatial Reasoning in Vision-Language Models via JEPA World Models

Sept 2025 – Dec 2025

- Designed and trained a Vision-Language Model (VLM) that replaces CLIP-style vision encoders with a self-supervised I-JEPA world model, investigating whether its structural representations improve spatial reasoning.
- Built a feature-translation module and two-stage alignment pipeline mapping I-JEPA's geometric visual features into the Qwen2-7B language embedding space for instruction-following multimodal reasoning, trained efficiently with LoRA and 4-bit quantization.

Autonomous Human Following Robot

Jan 2025 – May 2025

- Built and deployed a **real-time multi-sensor fusion perception and control system** using **RGB camera, depth sensors, and LiDAR** for autonomous human following on a **ROS-enabled omnidirectional robot**, converting visual tracking signals into closed-loop velocity commands.
- Developed a custom **SORT-based re-identification module** with lightweight appearance features to maintain target identity through occlusions, and engineered a **modular multi-node ROS architecture** optimized for **Jetson**, achieving **10–15 FPS** under embedded compute and latency constraints.

Non-Linear State Estimation with Square-Root UKF

Jan 2025 – May 2025

- Developed an end-to-end multi-camera vehicle tracking pipeline combining a fine-tuned object detector with a numerically stable Square-Root Unscented Kalman Filter (Sqrt-UKF) to handle low-light conditions, occlusions, intermittent sensor data, and non-linear vehicle dynamics.
- Derived the continuous-time linearization of a physics-based Bicycle Kinematics Motion Model, improving upon CV and CTRV models, and integrated the Bayesian filter and deep-learning detector into an end-to-end differentiable pipeline for joint optimization of motion-model and detector parameters.

Sabio Inc - Data Scientist

May 2022 – Mar 2023

- Developed audience segmentation and user-interest prediction systems using K-Means clustering, Logistic Regression, and XGBoost, delivering a **30% improvement in ad targeting precision**.
- Built and optimized automated AWS-based data pipelines with Apache Airflow, Spark, Scala, and Python, achieving **35% faster runtimes** and **25% lower compute costs**.